

SEMT20001: Principles of Computational Modelling

COURSEWORK ASSESSMENT: QUESTION 4

The question below forms one part of the assessment for the unit.

This is question 4 of 4.

Each question carries 25 marks. Answer all questions.

The maximum mark for this assessment is 100 marks.

The deadline for submission for all questions is **1pm on Thursday 19 March 2026**.

You must submit **two** items (on Blackboard) for **each** question:

- (1) a single self-contained Jupyter Notebook, that reports your answers to the question, and explains how you found them, including all relevant Python commands and functions, and explanations of your code design, **and**
- (2) a single .zip file containing all Python .py files.

Your Jupyter Notebook (1) for this question must contain everything you wish to be marked for this question (including, but not limited to, numerical and textual answers, figures, and all Python code used to generate them). Material not contained in your document (1) will receive no credit.

Any Python .py files that are not contained in your .zip file (2) will receive no credit.

Your submission must run from the pocm2526 environment, as defined on Blackboard (see *Learning and Teaching Area : Read me first! Python starter guide* for details). Any code that does not will receive no credit.

Your submission must be your own individual effort; submissions will be checked for possible plagiarism. You may only use tools such as spelling and grammar checkers in this assignment, and their use should be limited to corrections of your own work rather than substantial re-writes or extended contributions. Use of AI applications (even if they are free) beyond this is contract cheating; see <https://www.bristol.ac.uk/students/support/academic-advice/academic-integrity/contract-cheating/>. This is an AI Category 2 assessment; see <https://www.bristol.ac.uk/bilt/sharing-practice/guides/guidance-on-ai/using-ai-in-assessment/> for full details.

Question 4: Equation-based modelling

This part of Question 4 concerns the equation-based model given by the ODE problem

$$\frac{dy}{dt} = -\alpha(y - \cos(t)) - \sin(t), \quad y(0) = 1, \quad (1)$$

for $y = y(t)$, where α is a real constant.

- (a) Show (analytically) that the ODE (1) has solution $y(t) = \cos(t)$.

(1 mark)

- (b) Using the `myRK4` function in the lecture slides/on Blackboard, solve the ODE problem (1) with $\alpha = 1000$ for $t \in [0, \pi]$, using the *4th order Runge Kutta* method with stepsizes $h = \pi/4$, $h = \pi/16$ and $h = \pi/64$, and plot y vs. t graphs of the numerical solutions, together with the exact solution. What value of $h = \pi/2^n$ do you need to get a qualitatively correct solution to the ODE (1) with $\alpha = 1000$, using the 4th order Runge Kutta method?

Explain the results, in the light of the theory of numerical ODE solvers.

(5 marks)

- (c) The iteration

$$y_{k+1} = y_k + \frac{h}{2}f(t_k, y_k) + \frac{h}{2}f(t_{k+1}, y_{k+1}) \quad (2)$$

defines the *trapezoid* method to numerically solve the ODE system

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0.$$

Apply the trapezoid method (2) to the specific case of the ODE (1) and its right-hand-side function $f(t, y)$, and solve the resulting equation for y_{k+1} to give an explicit analytic expression

$$y_{k+1} = F(y_k, t_k, t_{k+1}), \quad y(0) = 1, \quad (3)$$

that could be used to solve the ODE (1) numerically.

Turn your iteration (3) into Python code, and use it to solve the ODE (1) with $\alpha = 100$, with stepsizes $h = \pi/4$, $h = \pi/16$ and $h = \pi/64$. Plot y vs. t graphs of the numerical solutions obtained with the trapezoid method, together with the exact solution, and comment on the results.

Plot a graph of the absolute error of the numerical solution of equation (1) found by the trapezoid method, versus step-size h , for a suitably-chosen range of h . Use the graph to find the order of the truncation error as $h \rightarrow 0$.

(7 marks)

- (d) Use the results of parts (b) and (c) to compare and contrast the strengths and weaknesses of the 4th order Runge Kutta and trapezoid methods.

(2 marks)

This part of Question 4 concerns an equation-based model of a susceptible-infected-susceptible (SIS) type disease on a network. In the model, individuals are divided into two groups: those who are susceptible to, and those who are infected with, a disease. Infection does not confer immunity; infected individuals return to the susceptible state upon recovery. Individuals are connected by

a set of bidirectional on/off links, and there are no self-links in the network. In addition to the usual infection and recovery processes, susceptible individuals can try to protect themselves from the disease by breaking a link with an infected neighbour and forming a new link with a susceptible individual instead. The dynamics of the underlying agent-based model are approximated¹ by the ODE system

$$\frac{dI}{dt} = pL_{SI} - rI, \quad (4)$$

$$\frac{dL_{II}}{dt} = pL_{SI} \left(\frac{L_{SI}}{S} + 1 \right) - 2rL_{II}, \quad (5)$$

$$\frac{dL_{SS}}{dt} = (r + w)L_{SI} - \frac{2pL_{SI}L_{SS}}{S}, \quad (6)$$

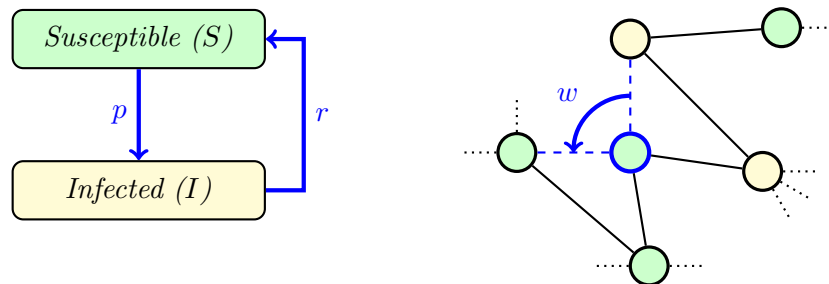
where the three variables (I, L_{II}, L_{SS}) represent

- $I = I(t)$, the proportion of infected individuals,
- $L_{II} = L_{II}(t)$, the density of links between two infected individuals in the network, and
- $L_{SS} = L_{SS}(t)$, the density of links between two susceptible individuals in the network,

as functions of time t . The total number of individuals is conserved, so the proportion of susceptible individuals $S = 1 - I$, and the total number of links is conserved, so $L_{SS} + L_{II} + L_{SI} = \hat{K}$, where L_{SI} is the density of links between susceptible and infected individuals, and $\hat{K}$ is a constant representing the overall link density; in a finite network with N nodes and K links, $\hat{K} = K/N$. The three parameters, (p, r, w) represent the probabilities (per unit time) of the three processes taking place on the network:

- p , the probability (per SI link) that susceptible individuals become infected, as a result of contact with an infected neighbour,
- r , the probability that an infected individual recovers, returning to the susceptible state,
- w , the probability that a susceptible individual breaks a link with an infected neighbour, and replaces it with a new link to a randomly selected susceptible individual (thus replacing an SI link with a new SS link),

illustrated schematically below:



¹Gross, T., Dommar D'Lima, C. J., Blasius, B., Phys. Rev. Lett. **96**, 208701 (2006)

- (e) Use an in-built Python ODE solver to solve the SIS model (4)–(6), with parameter values $(p, r, w) = (0.0015, 0.002, 0.05)$, $\hat{k} = 10$ for $t \in [0, 1000]$, and two different initial conditions $(I(0), L_{II}(0), L_{SS}(0)) = (0.1, 0.1, 8.1)$ and $(I(0), L_{II}(0), L_{SS}(0)) = (0.9, 8.1, 0.1)$, and plot a graph of the solutions versus time. Explain the results you obtain, in the context of the infection and the network.

How long, in each case, does it take for the solution to reach a steady state, where the magnitude of the rate of change of the solution, $\left| \frac{d\mathbf{y}}{dt} \right|$, is less than 10^{-4} ?

Plot graphs of the *steady-state* solutions of the SIS model versus p , for $p \in [0, 0.01]$, by using an in-built Python ODE solver to solve the SIS model (4)–(6) until the dynamics reach a steady state for a range of different values of p . Fix $r = 0.02$ and $\hat{k} = 10$ as above, and choose at least three different values of w . Explain the results you obtain, in the context of the infection and the network.

(8 marks)

- (f) In what ways would you expect the results of the equation-based model to be similar to/different from those of a corresponding agent-based model of the same disease? Describe the pros and cons of the two approaches. [NB. You are not expected to construct or simulate the agent-based model.]

(2 marks)